

Lecture 08: Bias-Variance Tradeoff

Decomposing Out-of-Sample Error in Regression

Caltech CS156: Learning from Data

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1 Conceptual Framework

In VC analysis, out-of-sample error E_{out} is bounded via in-sample error E_{in} and model complexity penalty Ω . In contrast, the **Bias-Variance Tradeoff** provides an alternative, non-bounding analysis of E_{out} specifically suited for regression problems under Mean Squared Error (MSE).

2 Mathematical Derivation of Bias-Variance Decomposition

Let $f(x)$ be the deterministic target function and $g^{(D)}(x)$ be the hypothesis learned from a training dataset $\mathcal{D}$ of size N . The expected out-of-sample error over all possible datasets $\mathcal{D}$ is:

$$\mathbb{E}_{\mathcal{D}}[E_{\text{out}}(g^{(D)})] = \mathbb{E}_{\mathcal{D}} \left[\mathbb{E}_{\mathbf{x}} \left[\left(g^{(D)}(\mathbf{x}) - f(\mathbf{x}) \right)^2 \right] \right] = \mathbb{E}_{\mathbf{x}} \left[\mathbb{E}_{\mathcal{D}} \left[\left(g^{(D)}(\mathbf{x}) - f(\mathbf{x}) \right)^2 \right] \right] \quad (1)$$

Define the **average hypothesis** $\bar{g}(\mathbf{x})$ over all datasets:

$$\bar{g}(\mathbf{x}) = \mathbb{E}_{\mathcal{D}} \left[g^{(D)}(\mathbf{x}) \right] \quad (2)$$

Adding and subtracting $\bar{g}(\mathbf{x})$ inside the expectation:

$$\mathbb{E}_{\mathcal{D}} \left[\left(g^{(D)}(\mathbf{x}) - f(\mathbf{x}) \right)^2 \right] = \mathbb{E}_{\mathcal{D}} \left[\left(\left(g^{(D)}(\mathbf{x}) - \bar{g}(\mathbf{x}) \right) + \left(\bar{g}(\mathbf{x}) - f(\mathbf{x}) \right) \right)^2 \right] \quad (3)$$

$$\begin{aligned} &= \mathbb{E}_{\mathcal{D}} \left[\left(g^{(D)}(\mathbf{x}) - \bar{g}(\mathbf{x}) \right)^2 \right] + \left(\bar{g}(\mathbf{x}) - f(\mathbf{x}) \right)^2 \\ &\quad + 2 \left(\bar{g}(\mathbf{x}) - f(\mathbf{x}) \right) \underbrace{\mathbb{E}_{\mathcal{D}} \left[g^{(D)}(\mathbf{x}) - \bar{g}(\mathbf{x}) \right]}_{=0} \end{aligned} \quad (4)$$

Thus, the cross-term vanishes completely, leaving two distinct components:

Bias-Variance Decomposition Formula

$$\mathbb{E}_{\mathcal{D}}[E_{\text{out}}(g^{(D)})] = \text{Bias} + \text{Variance} \quad (5)$$

where:

$$\text{Bias}(\mathbf{x}) = (\bar{g}(\mathbf{x}) - f(\mathbf{x}))^2 \implies \text{Bias} = \mathbb{E}_{\mathbf{x}} \left[(\bar{g}(\mathbf{x}) - f(\mathbf{x}))^2 \right] \quad (6)$$

$$\text{Var}(\mathbf{x}) = \mathbb{E}_{\mathcal{D}} \left[\left(g^{(D)}(\mathbf{x}) - \bar{g}(\mathbf{x}) \right)^2 \right] \implies \text{Variance} = \mathbb{E}_{\mathbf{x}} \left[\mathbb{E}_{\mathcal{D}} \left[\left(g^{(D)}(\mathbf{x}) - \bar{g}(\mathbf{x}) \right)^2 \right] \right] \quad (7)$$

- **Bias**: Measures the inherent limitation of the hypothesis space $\mathcal{H}$ to approximate $f(x)$ even with infinite data.
- **Variance**: Measures the sensitivity/instability of the learned hypothesis to the specific choice of dataset $\mathcal{D}$.

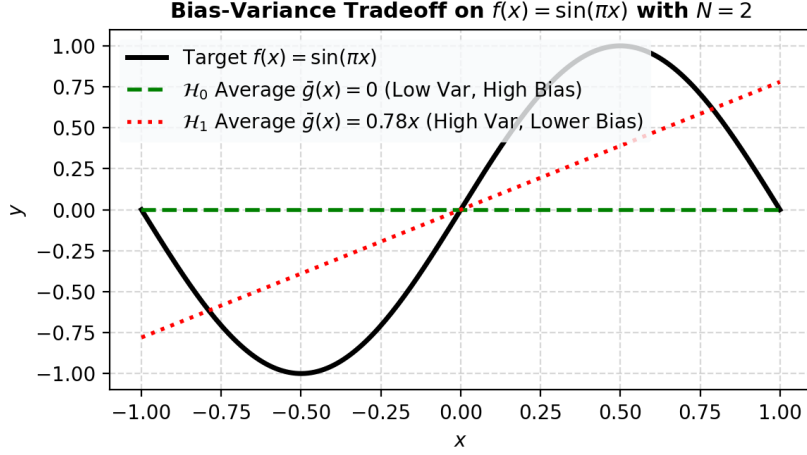


Figure 1: Model $\mathcal{H}_0$ (flat line) vs $\mathcal{H}_1$ (sloped line) trained on $N = 2$ points from $\sin(\pi x)$.

Table 1: Bias-Variance Comparison on $f(x) = \sin(\pi x)$ with $N = 2$

Model	Bias	Variance	Total E_{out}
$\mathcal{H}_0 : h(x) = b$	0.50	0.25	0.75
$\mathcal{H}_1 : h(x) = ax + b$	0.21	1.69	1.90

3 Case Study: Sine Wave Approximation with $N = 2$

Consider target $f(x) = \sin(\pi x)$ on $x \in [-1, 1]$. We receive only $N = 2$ training points. Compare two models:

- $\mathcal{H}_0$: Horizontal lines $h(x) = b$.
- $\mathcal{H}_1$: Lines passing through two points $h(x) = ax + b$.

Takeaway: Although $\mathcal{H}_1$ has lower bias ($0.21 < 0.50$), its variance exploded (1.69) because $N = 2$ points cause massive line slope fluctuations. Thus, the simpler model $\mathcal{H}_0$ yields far lower out-of-sample error!

4 Analytic Learning Curves for Linear Regression

When learning a target with inherent noise σ^2 using Linear Regression with d features:

$$\mathbb{E}_{\mathcal{D}}[E_{\text{in}}] = \sigma^2 \left(1 - \frac{d+1}{N} \right) \quad (8)$$

$$\mathbb{E}_{\mathcal{D}}[E_{\text{out}}] = \sigma^2 \left(1 + \frac{d+1}{N} \right) \quad (9)$$

As $N \rightarrow \infty$, both E_{in} and E_{out} converge to the noise floor σ^2 . The generalization gap is approximately $2\sigma^2 \frac{d+1}{N}$.